University of Science and Technology of Oran – Mohamed Boudiaf (USTOMB)
Faculty of Mathematics and Informatics
Computer Science Department

Master's Degree Courses

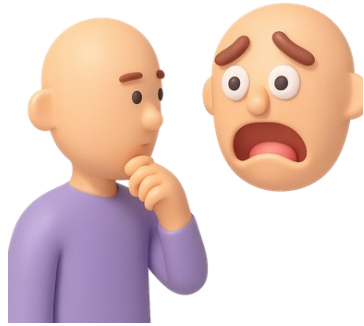
Artificial Intelligence and Applications

Animated Multimedia Applications Design

Facial Animation

ebarki informatics

Wye animating faces ?



- ✓ How ?
- ✓ Reference –
  - ➢ The human face



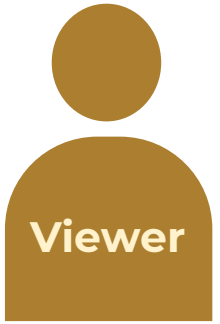






Emotional Expressions

Speech





Facial movements



Mandible movements

- ❑ **Vertical** – Lowering/raising: opening/closing mouth
- ❑ **Anteroposterior** – Front & rear chin projection.
- ❑ **Lateral** – Carrying the chin to the left & to the right



Hair, folds, wrinkles, colors

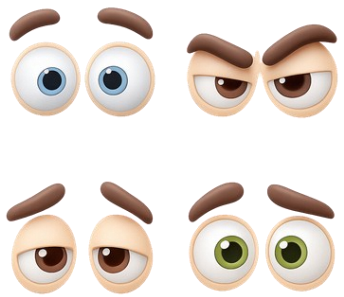
- ❑ Character traits
- ❑ Hair & hair elements: beard, mustaches …



Muscles

- ❑ Push/pull skin in different ways
- ❑ Direct in creation of expressions
- ❑ Chewing, the lips/eyes movements

Eyes & eyebrows



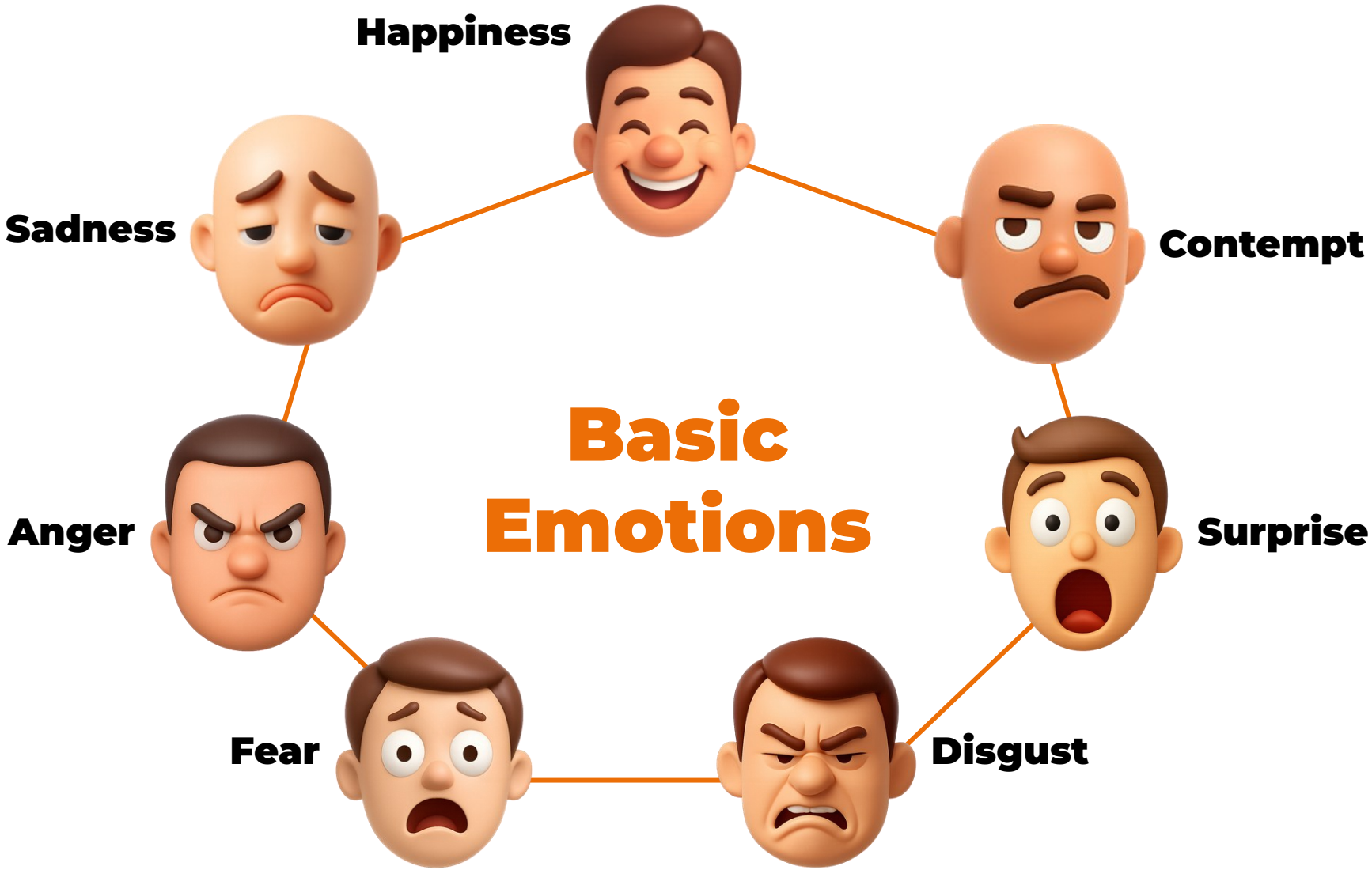
- ❑ Convey emotions & create expressions
- ❑ **Animation constraint –** Ensure convergence of directions of gazes pointed by the two eyes
- ❑ **Animations** – Pupil dilation, Iris coloring, Light reflection
- ❑ Accompany with eyebrow movements

The mouth



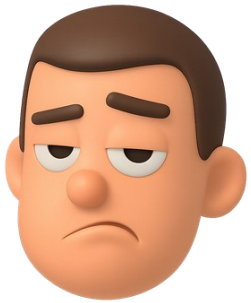
- ❑ Independent lower/upper lips
- ❑ Words, emotions
- ❑ Gums, teeth & tongue contribute to expression when visible in facial posture (secondary in animations but provide more realism).

Emotions

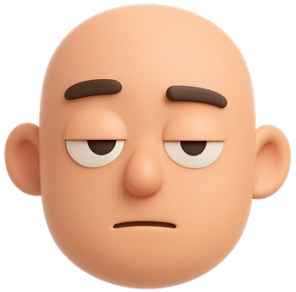

Happiness
Sadness
Contempt
Basic Emotions
Anger
Surprise
Fear
Disgust

Emotions

Secondary Emotions



Disdain



Disinterest



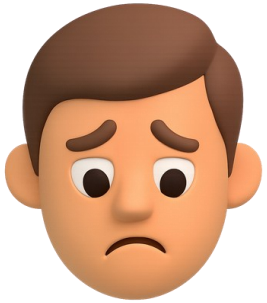
Deviousness



enthusiasm



Confusion



Worry



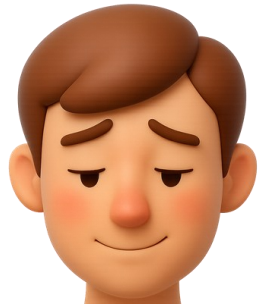
Terror



Rage



Weariness



shy

Realistic vs Symbolic

Realistic Animations

High-Fidelity Simulation of Human Anatomy

- → **Detailed anatomical structure** (bones, muscles, skin layers, soft-tissue dynamics)
- → **Biomechanical accuracy** – Follow natural constraints : muscle contraction, skin elasticity.
- → **High computational precision –** Requires advanced algorithms, motion capture data, physical simulation

Example Application –
→ Medical Training



Common Techniques

- → Parameterized Meshes
- → Analytical Vector Models

Symbolic Animations



Simplified, Expressive, Interpretable

- → **Simplified facial components**
  Reduced to geometric shapes: Circles, arcs, lines...
- → **Stylized expressions –** Clarity & emotional readability rather than anatomy
- → **Intuitive, exaggerated symbols** that audiences interpret instantly
- → **Low computational cost –** Suitable for mobile apps, stylized games, data vis., UI and pedagogical content.

Typical Elements

Eyes – Circles, dots, lines for emotion
Mouth – Arcs indicating emotions
Colors – Symbolically to reinforce mood

Enhanced by Audio

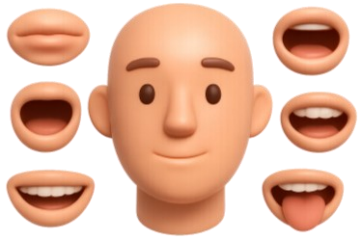
- → Voice tone, Rhythm, pauses, stress, prosody
- → Compensate for simplicity of the visual expression by adding emotional depth.

Visemes

Visual Counterpart of Speech Sounds

→ **Visual representations of phonemes**

→ **Facial movements & mouth shapes** required to simulate speech in animation



Importance of Visemes in Animation

- **Lip-sync –** films, games, VR, avatars
- **Speech-driven** facial animations
- Human–computer interaction, including **conversational agents**
- **Multimodal communication research** integrating audio & visual cues

Key Facial Components

Mouth Shape – Primary component – Speech animation

- Degree of **opening | Rounding** or **spreading** of the lips
- Vertical/horizontal deformations

Jaw Position – Particularly for open vowels

→ **Lowering** – Open vowels (/a/) | **Lifting** – Closed/semi-cl. (/i/, /u/)

Tongue Position – Accuracy/realism [Approximated otherwise]

→ **Front, central, or back vowels**.

→ Shapes consonants like /l/, /t/, /d/, /r/.

→ Enhances credibility of speech animation

Lips and Lip Corners – Lip tension and lip-corner

→ Rounded vowels (/o/, /u/) | Bilabial consonants (/p/, /b/, /m/)

→ Fricatives (/f/, /v/) – Lower lip touches upper teeth

Secondary Facial Movements

→ Cheek tension during lip stretching

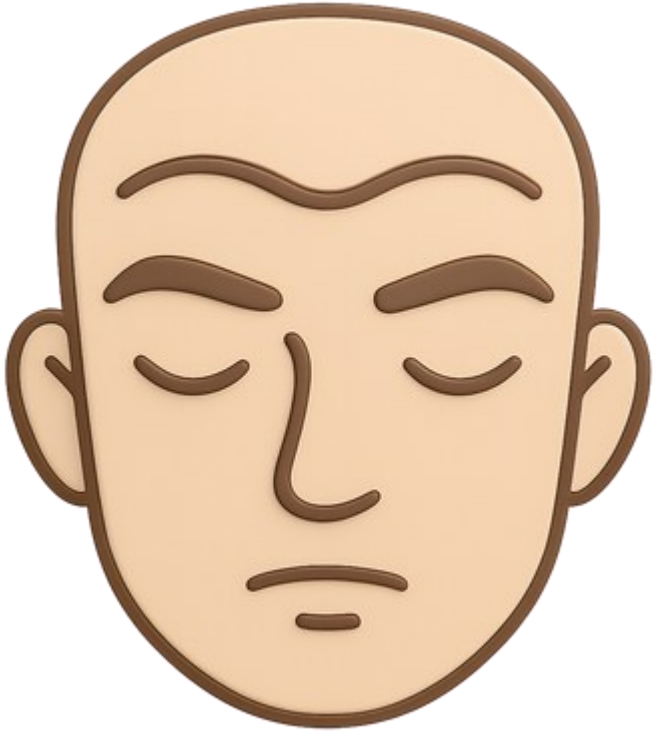
→ Subtle eyebrow movements

→ Micro expressions that occur naturally when talking

Facial Action Coding System (FACS)

- 👀 Anatomically based system describing observable facial movements
- 💪 Expressions → Action Units (AUs) – individual muscle movements
- 👨‍🔬 **Paul Ekman** & **Wallace V. Friesen**
- 🕵️‍♂️ Used in **psychology**, **emotion research**, **deception detection**, & **HCI**
- 🧐 Enables **precise analysis** of **emotions** and **nonverbal cues**

AU1 – Inner Brow Raiser



Application of FACS

🧠 **Scientific research** – Study emotions, nonverbal communication, & social interactions

🎨 **Animators** & computer scientists → Facial animation & emotion recognition

👀 Awareness of subtle **facial behaviors**

👩‍🏫 Psychotherapists, interviewers, educators, & communication specialists

FACS Manual

👀 **Criteria** to observe and code each Action Unit (AU)

🔄 **Explains** how AUs combine for complex expressions

👨‍🔬 First published in **1978** by **Paul Ekman** & **Wallace V. Friesen**

🔄 Most recently revised in **2002**

💼 Available from the Paul Ekman Group for **professional** and **academic** use

Codes for action units

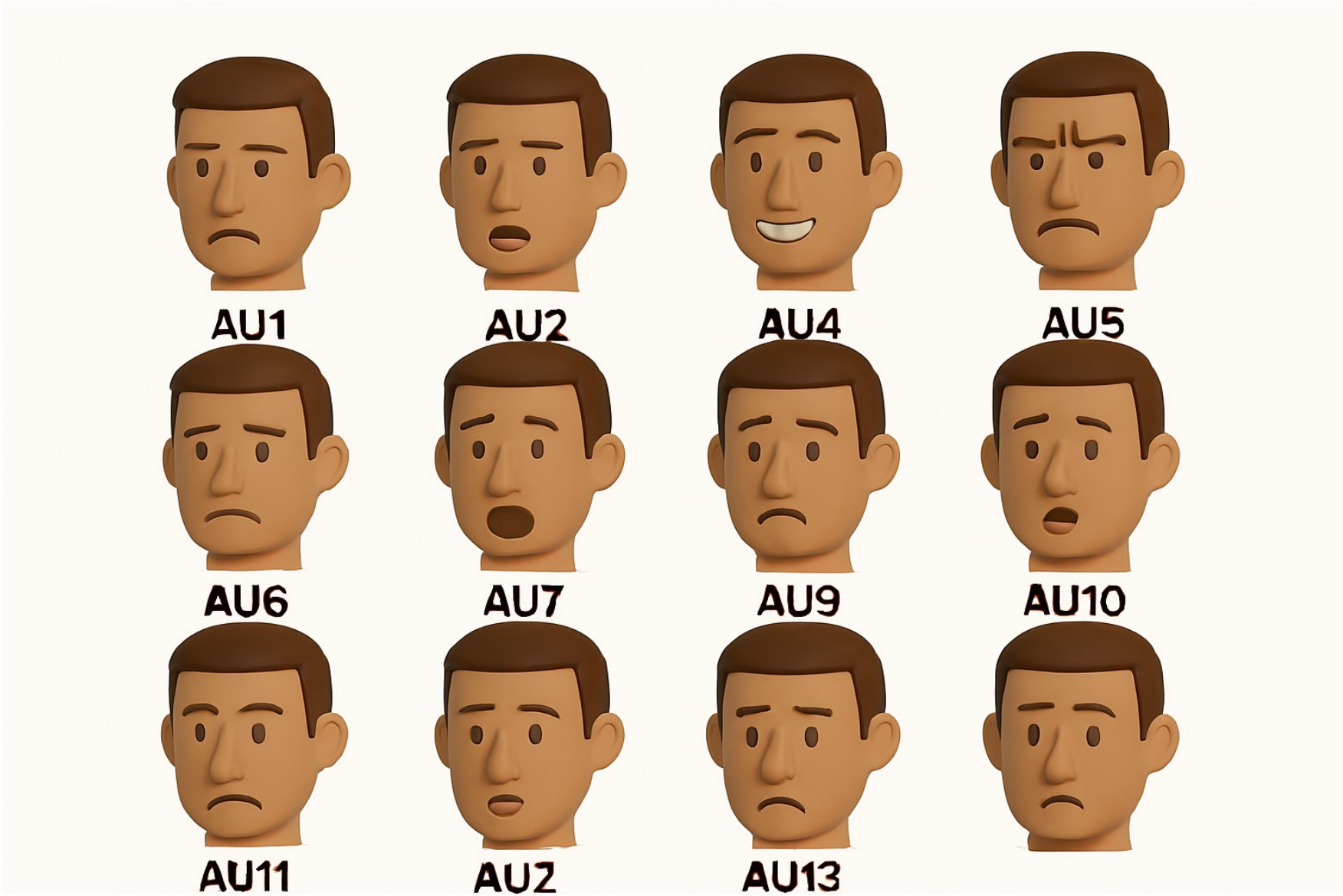
AUs – Fundamental actions of individual or group of muscles

Intensity scoring

- **A** – Trace
- **B** – Slight
- **C** – Marked or pronounced
- **D** – Severe or extreme
- **E** – Maximum

Face side

- **R** – On the right side of the face and L : on the left.
- **U** – Unilateral unit (only one side) without specific side
- **A** – An action which is unilateral but has a stronger side


AU1
AU2
AU4
AU5
AU6
AU7
AU9
AU10
AU11
AU2
AU13

Universal Emotions

- ❑ 6+12 ❑ Happiness



- ❑ 1+4+15 ❑ Sadness



- ❑ 1+2+5B+26 ❑ Surprise



- ❑ 1+2+4+5+7+20+26 ❑ Fear



- ❑ 4+5+7+23 ❑ Anger



- ❑ 9+15+17 ❑ Disgust

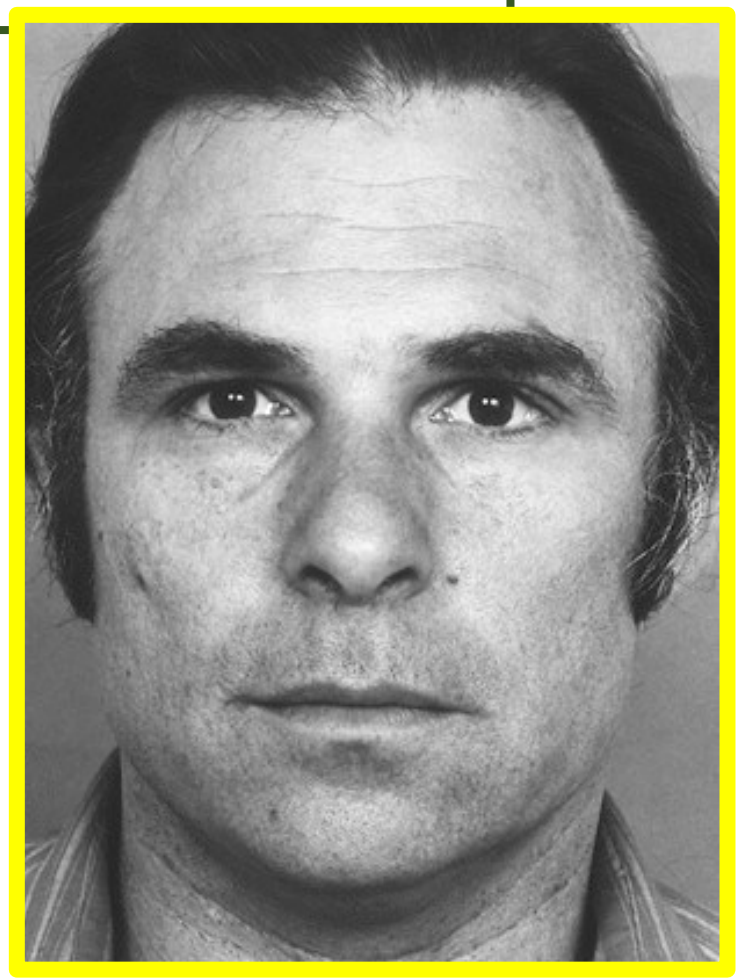


- ❑ R12A+R14A ❑ Contempt



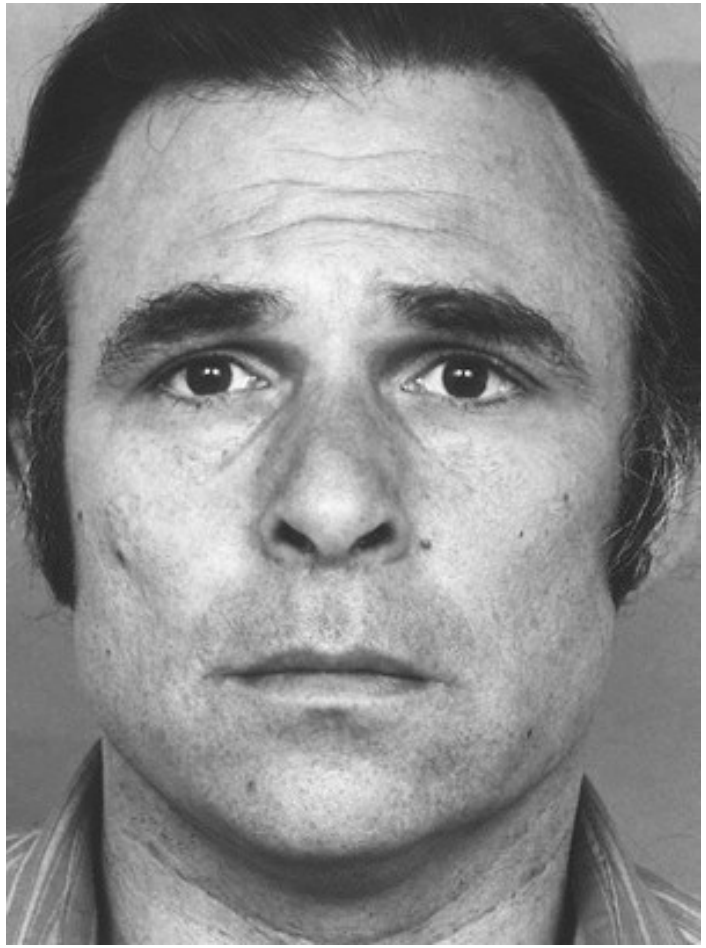
AU 0

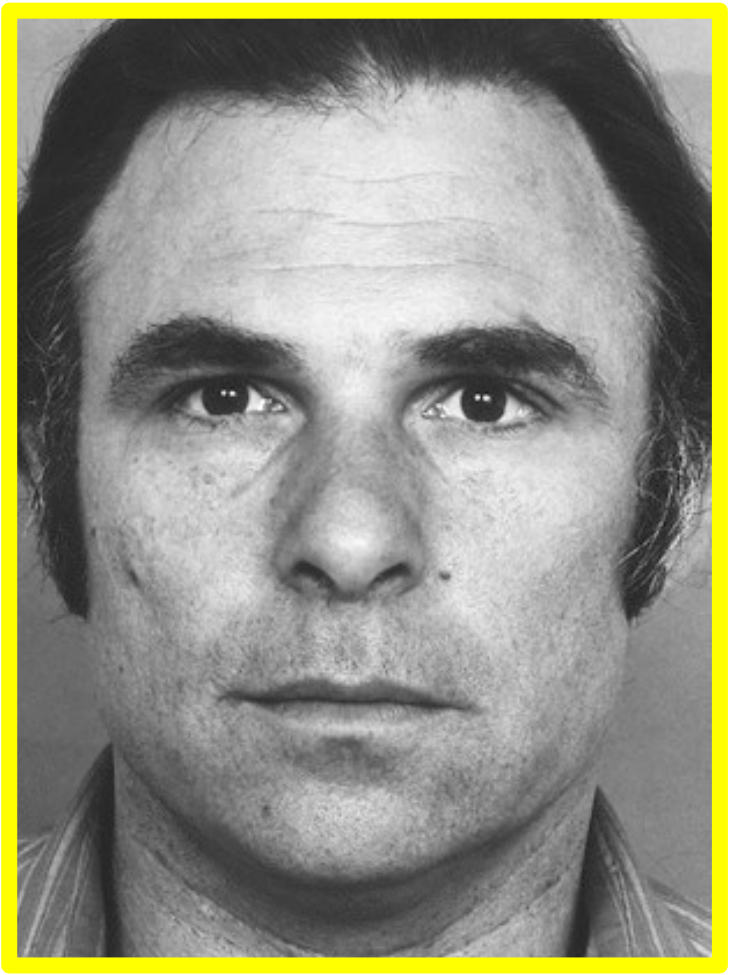
- The face is not showing any action, often called a neutral face.
- The face is not actually at rest because the eyes are open, the jaw is closed, but no AU can be scored.



AU 1

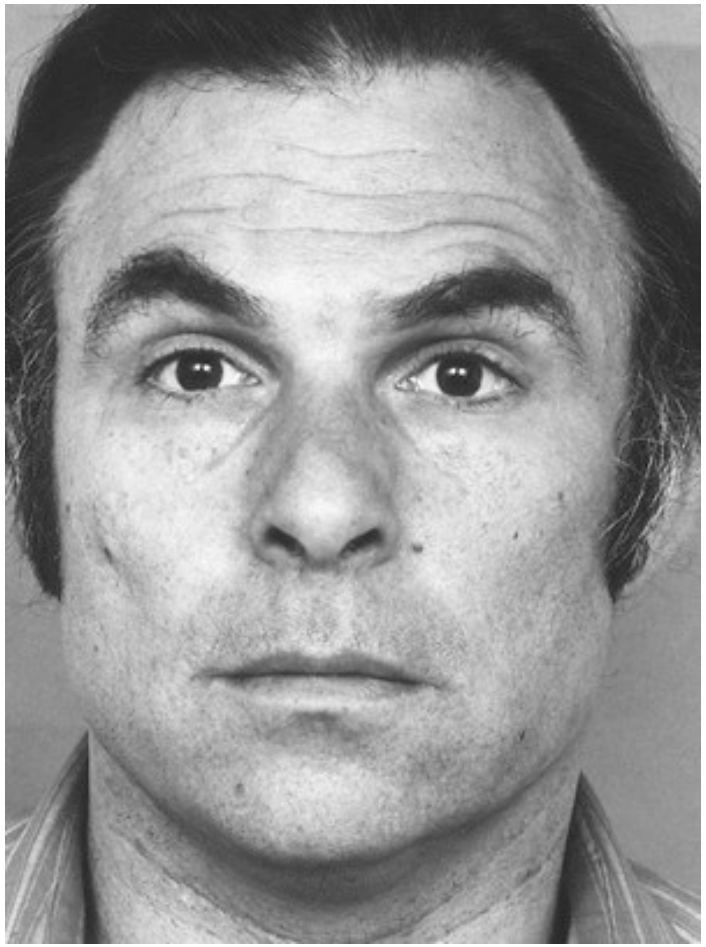
- The inner corners of the eyebrows are lifted slightly, the skin of the glabella and forehead above it is lifted slightly and wrinkles deepen slightly and a trace of new ones form in the center of the forehead.

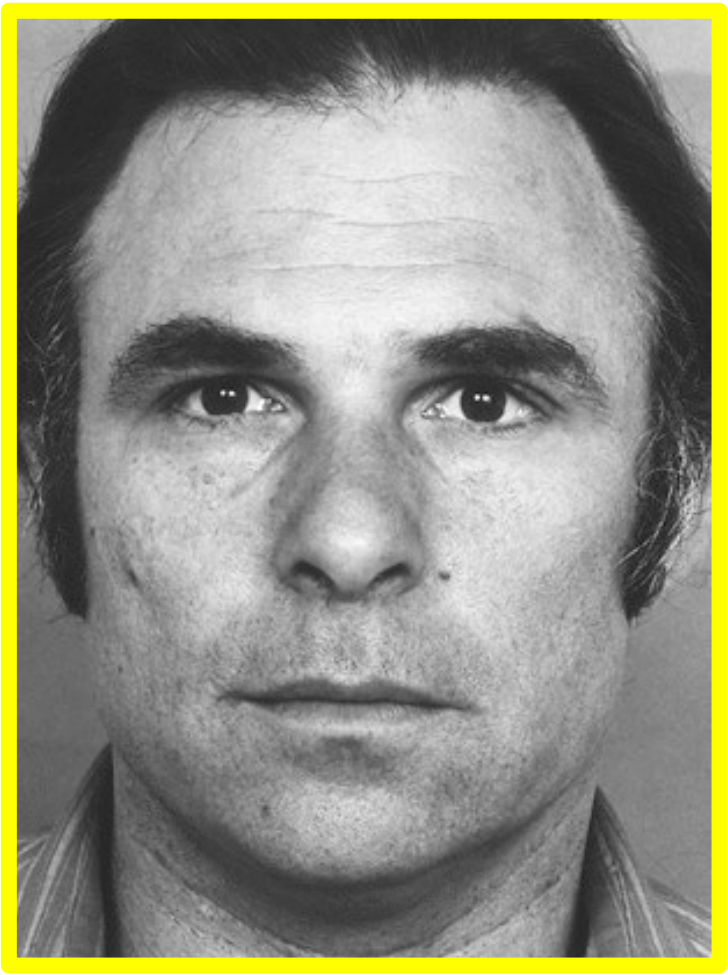




AU 2

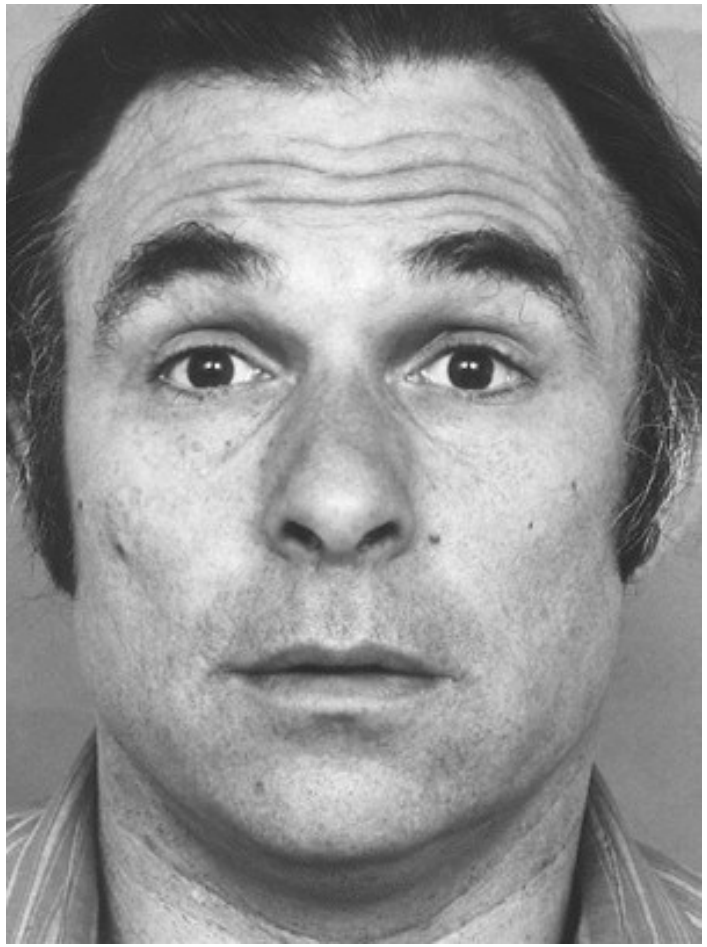
- The outer part of the eyebrow raise is pronounced. The wrinkling above the right outer eyebrow has increased markedly, and the wrinkling on the left is pronounced. Increased exposure of the eye cover fold and skin is pronounced.

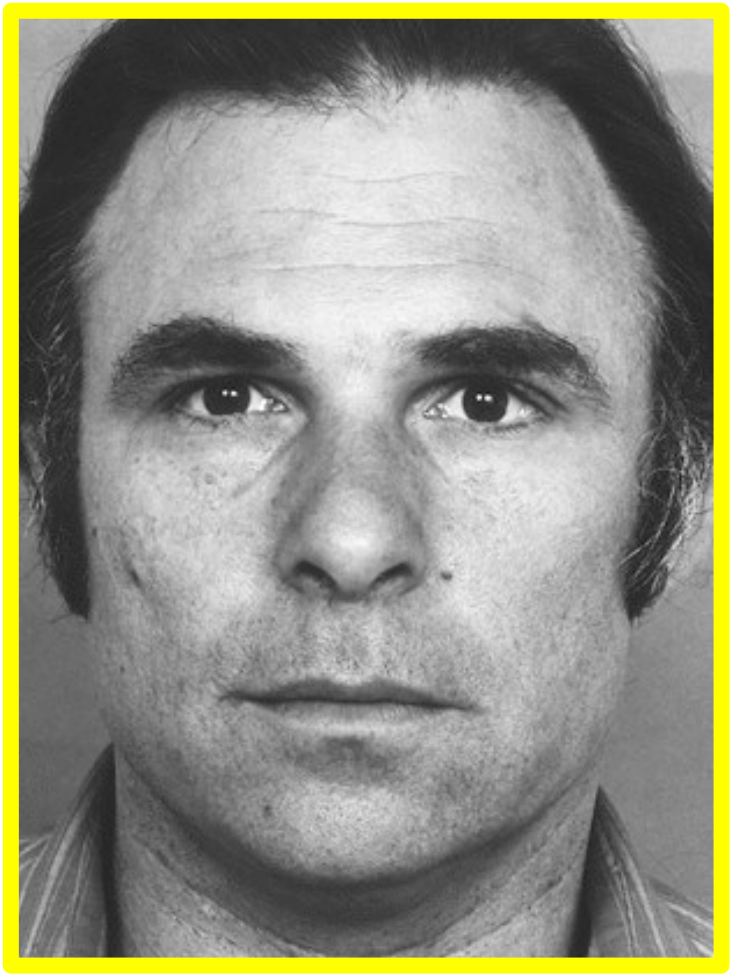




AU 1+2

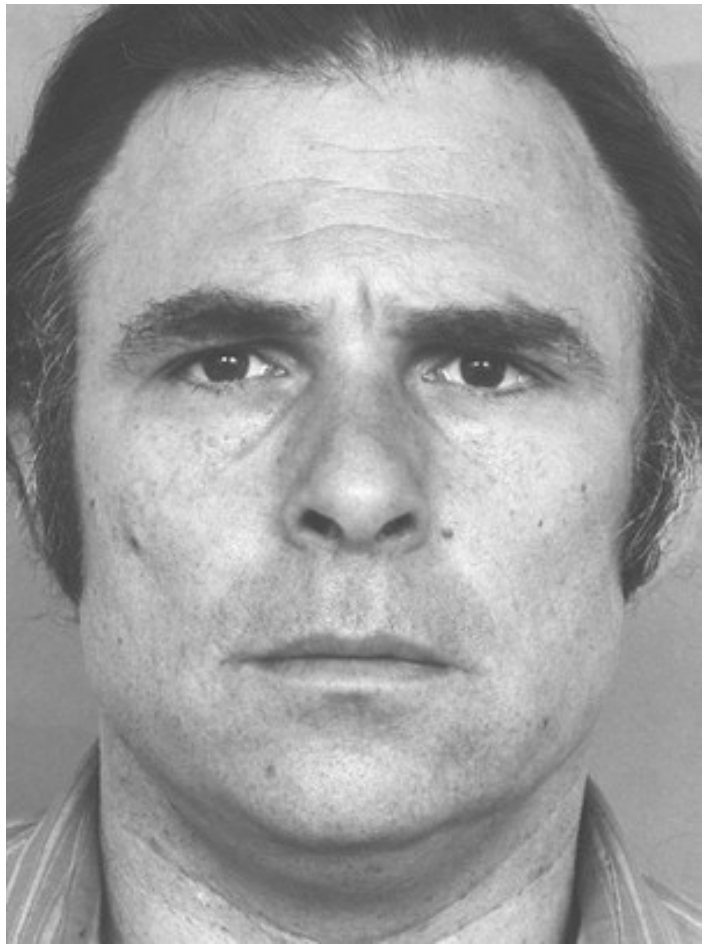
- This action is asymmetrical. The entire eyebrow is raised up extremely. The eye cover fold is exposed extremely.

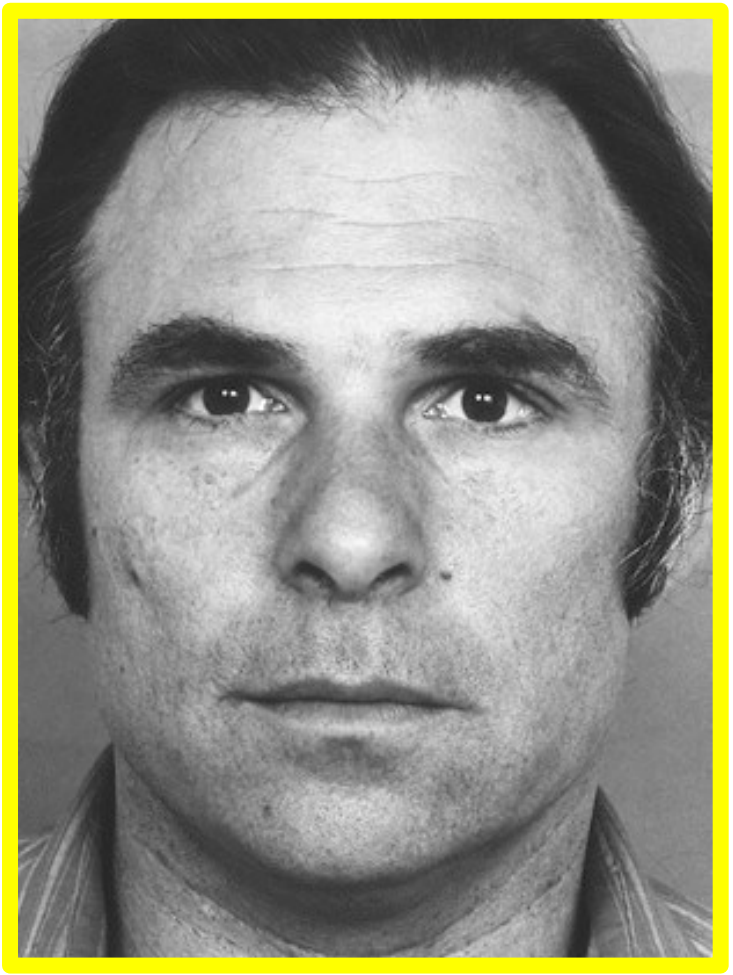




AU 4

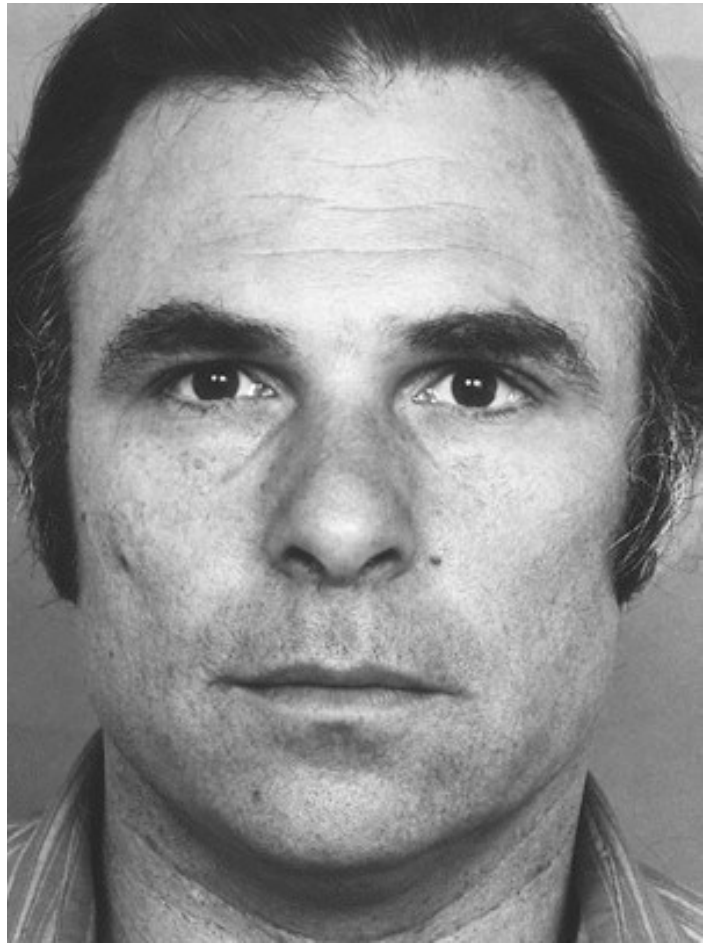
- Vertical wrinkles appear in the glabella and the eyebrows are pulled together.
- The inner parts of the eye-brows are pulled down a trace on the right and slightly on the left with traces of wrinkling at the corners.

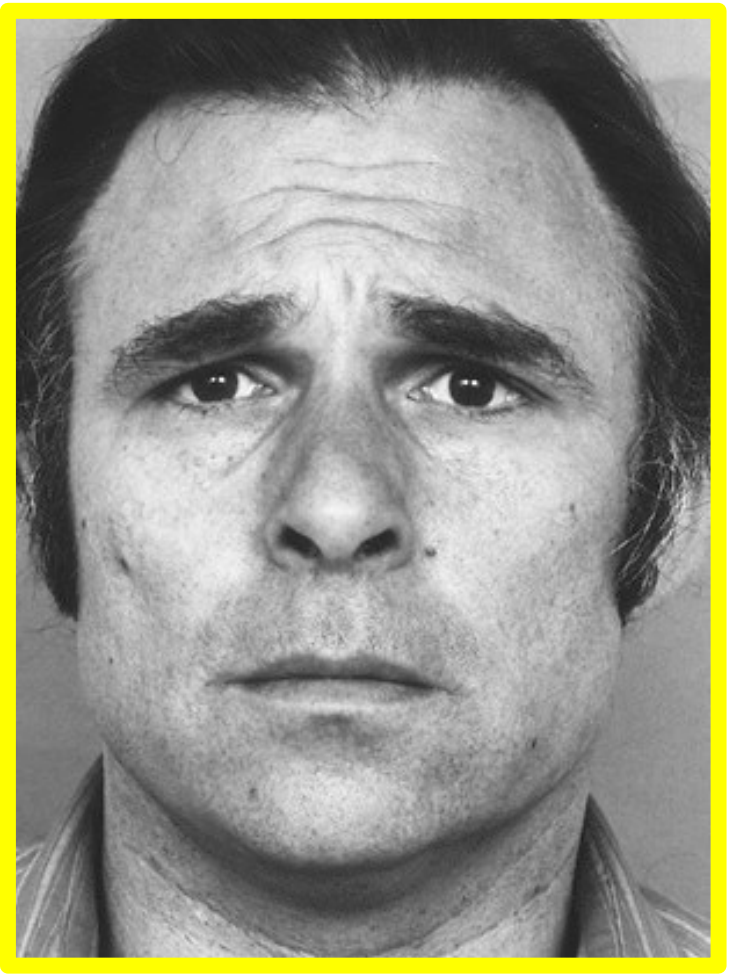




AU 1+4

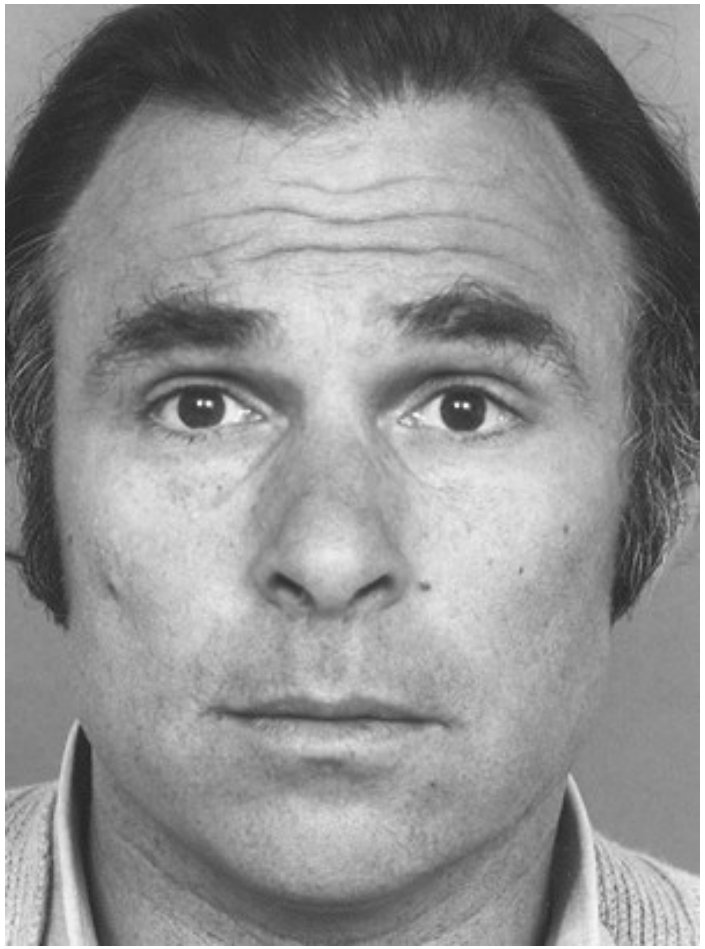
- Raising the inner corners of the eyebrows and pull the eyebrows together.

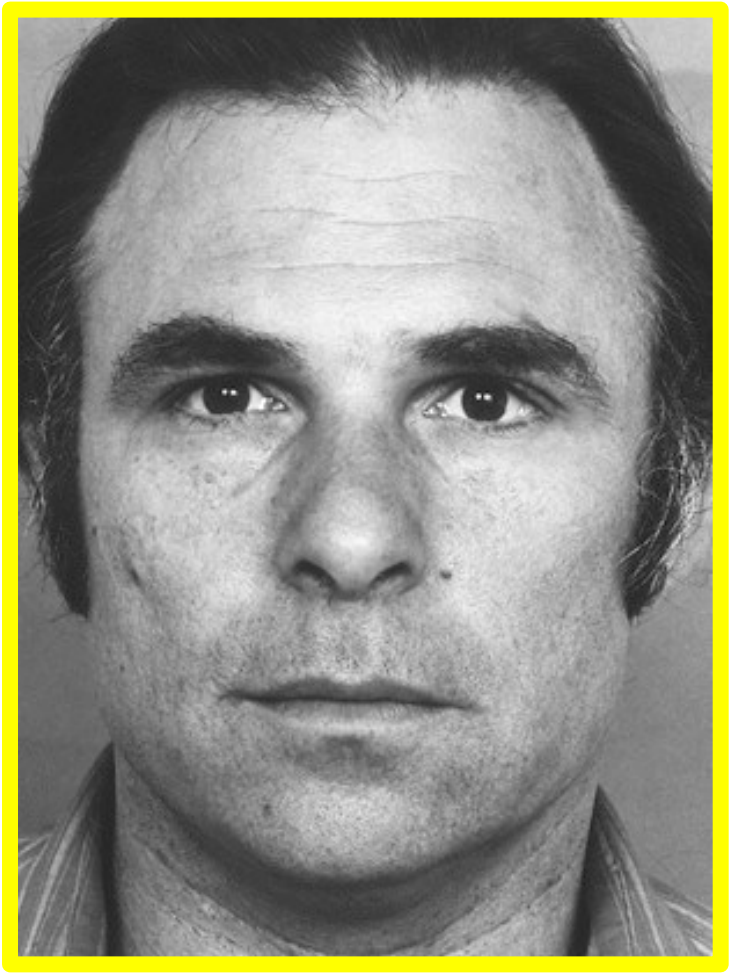




AU 1+2+4

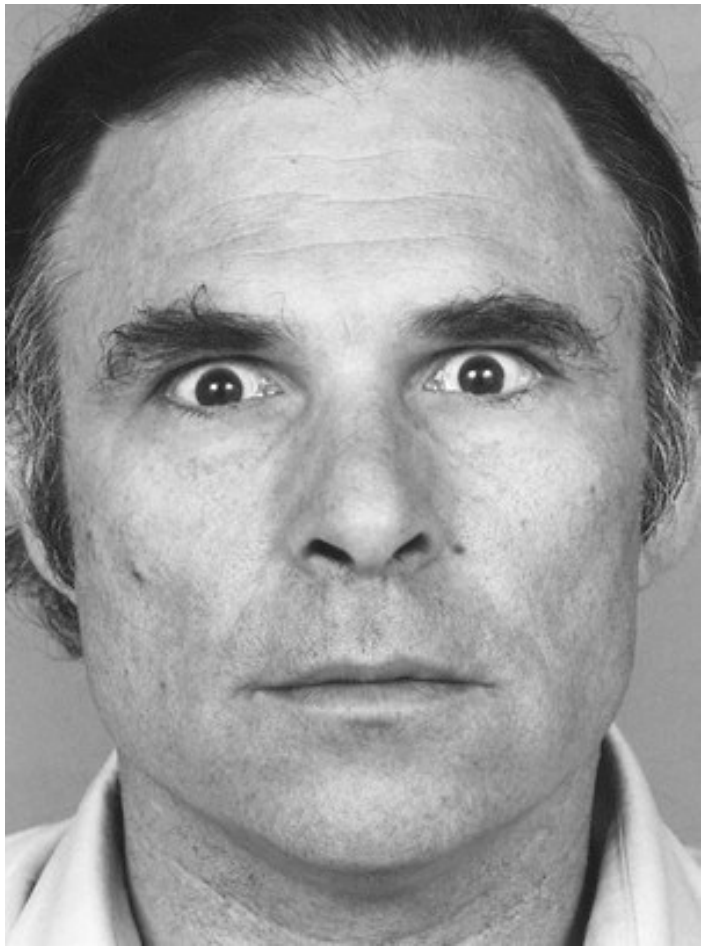
- The entire brow is raised markedly. The slight pulling together appears weaker than the raising, with only slight straightening and a trace of bulging between the brows.

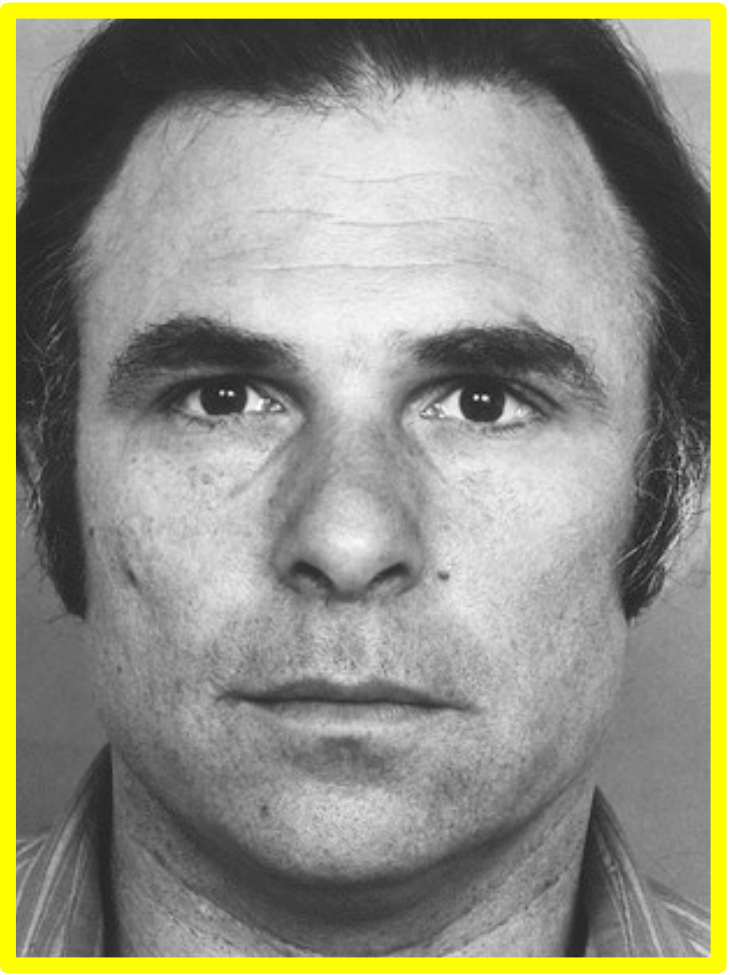




AU 5

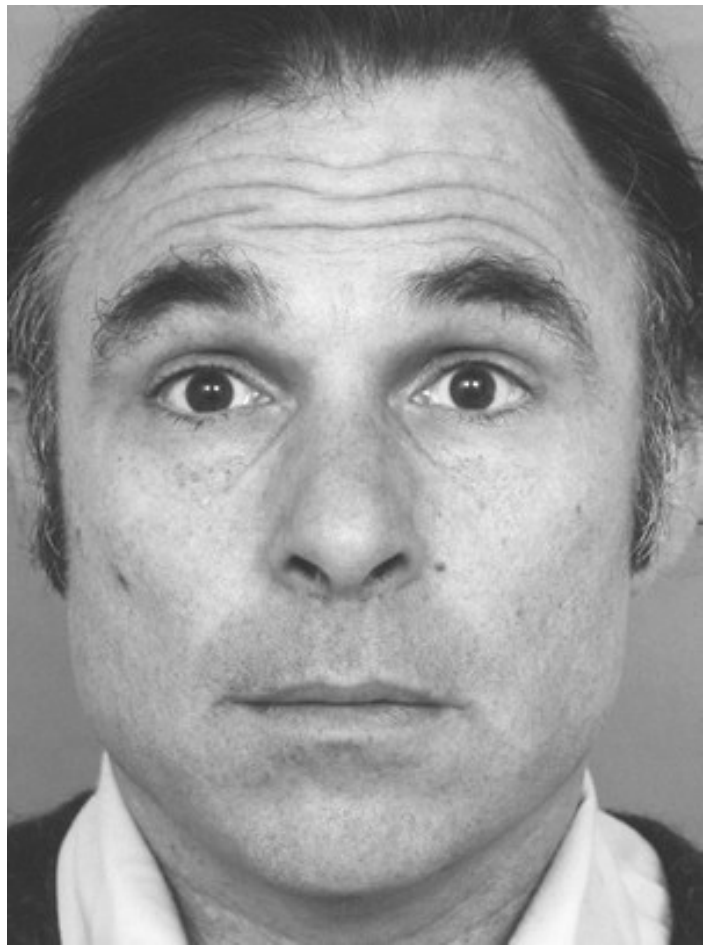
- The upper eyelid is pulled up, the medial part virtually disappearing into the revulsion margin of the eye cover fold. More sclera around the upper part of the iris is revealed
- The inner corner of the eye shows more mucosa and sclera

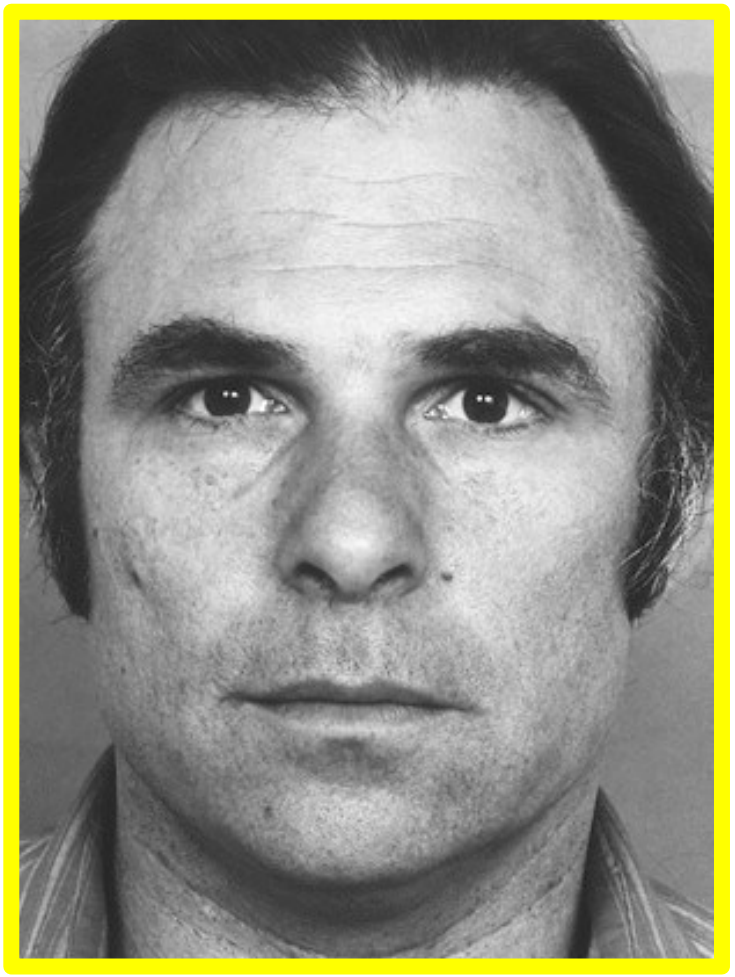




AU 1+2+5

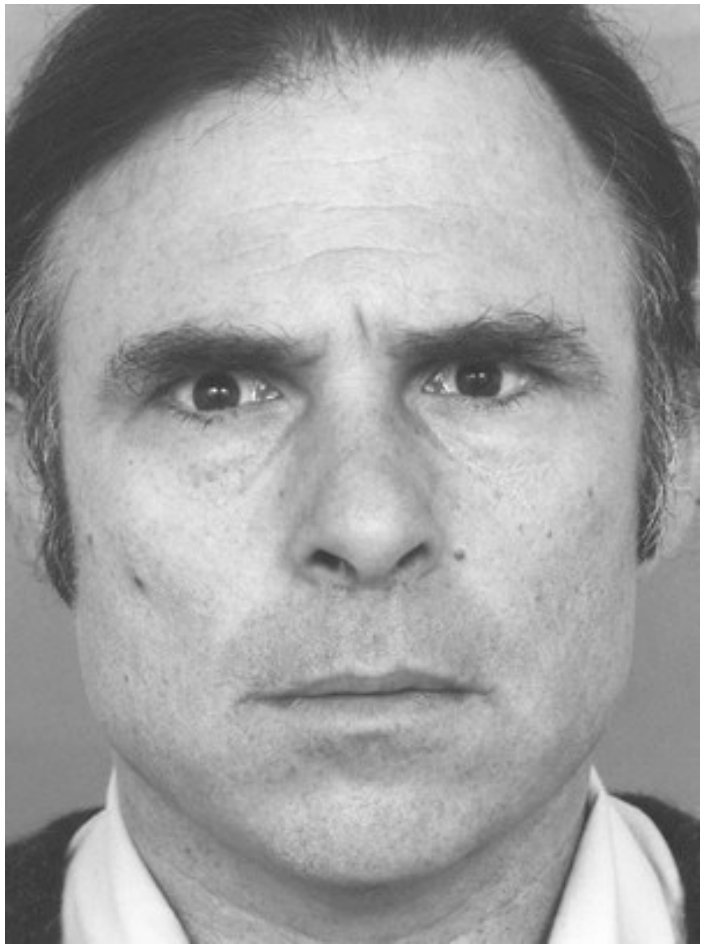
- Eye brows raised to expose the eye cover fold. The upper eyelid is raised to expose virtually the entire iris.

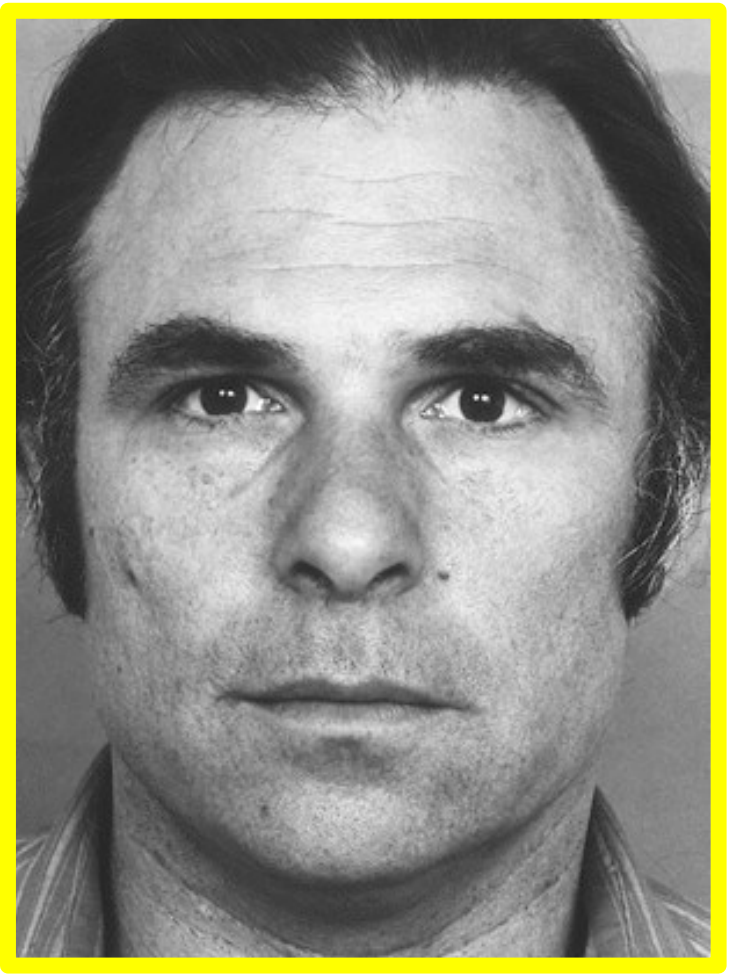




AU 4+5

- Pull the inner parts ofthe eyebrows are pulled together and raise the upper eyelid.





AU 1+2+4+5

